USB Insider Forensics: Detecting Data Exfiltration via Removable Media on Windows

Case: INSIDER-USB-001
Investigator: Student Investigator

Executive Summary

This controlled investigation evaluated device, filesystem, Registry/event, and user-activity o bservations. H3 is **SUPPORTED** (HIGH). The assessment relies on corroboration rather than a s ingle artifact.

Investigation and Research Questions

Did employee01 copy confidential company data to an external USB device, when, which files, and what independent evidence supports the conclusion? Can Windows artifacts distinguish legitimat e USB use from simulated insider leakage?

Scope and Evidence Sources

Offline analysis of Windows 10/11 artifact exports: a controlled two-scenario dataset using fic tional Example Corporation data, optionally complemented by read-only live triage of a real PC.

Evidence Inventory

- EVD-DEMO-001 | sample (artifact export bundle) | SHA-256 `1ee1023f9cc8f61f9c9cef5ff7679569479 9c3e1f7fd60296a04626dc8d6644e` | 5251 bytes
- Synthetic demonstration export bundle; not an acquired disk image and not real case evidence.

Raw exports are preserved separately; normalized records retain their original payload and prov enance identifiers.

Acquisition Methodology and Hash Verification

The included demo hashes the synthetic export bundle for provenance. Real evidence is either co llected read-only with the triage collector, which records a SHA-256 manifest, or acquired as a powered-off disk and USB image that is hashed, logged in custody and examined only as a verifi ed copy.

Artifact, USB, User, and Filesystem Analysis

- META-A: simulation_metadata / None / n/a (LOW)

- A-001: user_session / logged_on / n/a (MEDIUM)

- A-002: usb / connected / BACKUP_DRIVE (MEDIUM)

- A-003: filesystem / copied / Readme.txt (MEDIUM)

- A-004: usb / disconnected / BACKUP_DRIVE (MEDIUM)

- META-B: simulation_metadata / None / n/a (LOW)

- B-001: user_session / logged_on / n/a (MEDIUM)

- B-002: usb / connected / BACKUP_DRIVE (MEDIUM)

- B-003: lnk / accessed / Employee_Salaries.xlsx (MEDIUM)

- B-004: office / accessed / Customer_Database.csv (MEDIUM)

- B-005: filesystem / copied / Employee_Salaries.xlsx (MEDIUM)

- B-006: filesystem / copied / Customer_Database.csv (MEDIUM)

- B-007: archive / archived / Project_Files.zip (MEDIUM)

- B-008: recycle_bin / deleted / Project_Apollo.pdf (MEDIUM)

- B-009: usb / disconnected / BACKUP_DRIVE (MEDIUM)

Unified Timeline

- 2026-09-15T09:00:00+05:30 — A-001 — Interactive laboratory user session

- 2026-09-15T09:03:00+05:30 — A-002 — Removable storage device observed

- 2026-09-15T09:06:00+05:30 — A-003 — Normal test file created on lab USB

- 2026-09-15T09:10:00+05:30 — A-004 — Device-removal observation

- 2026-09-15T10:00:00+05:30 — B-001 — Interactive laboratory user session

- 2026-09-15T10:04:00+05:30 — B-002 — Registry device identity observation

- 2026-09-15T10:07:00+05:30 — B-003 — Shortcut artifact associated with classified file

- 2026-09-15T10:08:00+05:30 — B-004 — Recent-document observation

- 2026-09-15T10:10:00+05:30 — B-005 — Destination-side file entry on laboratory USB

- 2026-09-15T10:11:00+05:30 — B-006 — Destination-side file entry on laboratory USB

- 2026-09-15T10:13:00+05:30 — B-007 — ZIP archive created during device session

- 2026-09-15T10:14:00+05:30 — B-008 — Temporary classified-file copy deletion observation

- 2026-09-15T10:18:00+05:30 — B-009 — Device-removal observation

- unknown — META-A —

- unknown — META-B —

Correlations

- CORR-001 USER_SESSION (MEDIUM): User-session artifact observed

- CORR-002 USB_SESSION (HIGH): Removable-device connection artifact observed; Disconnect artifa
ct observed; session has a start and an end

- CORR-003 USER_SESSION (MEDIUM): User-session artifact observed

- CORR-004 USB_SESSION (HIGH): Removable-device connection artifact observed; Disconnect artifa
ct observed; session has a start and an end

- CORR-005 POSSIBLE_TRANSFER_SESSION (HIGH): USB device activity observed; Sensitive-file act
ity occurred within the configured time window; Employee user-session artifact corroborates tem
poral association; Destination-side file artifact was observed on removable media; Archive acti
vity occurred in the same experiment

- CORR-006 SENSITIVE_FILE_ACTIVITY (MEDIUM): Project-classified sensitive files have activity a
rtifacts

- CORR-007 ARCHIVE_ACTIVITY (MEDIUM): Archive creation artifact observed; intent is not establi
shed

- CORR-008 POSSIBLE_CONCEALMENT (LOW): Rename or deletion artifact observed; Artifact presen
alone does not establish malicious intent

Hypothesis Evaluation

H1 — SUPPORTED (HIGH)
- USB connection artifacts support device presence

H2 — SUPPORTED (MEDIUM)
- User-associated sensitive-file artifacts were observed

H3 — SUPPORTED (HIGH)
- USB device activity observed
- Sensitive-file activity occurred within the configured time window
- Employee user-session artifact corroborates temporal association
- Destination-side file artifact was observed on removable media
- Archive activity occurred in the same experiment

H4 — PARTIALLY_SUPPORTED (MEDIUM)
- Rename or deletion activity is consistent with possible concealment
- Artifact presence alone does not establish malicious intent

Observation, Inference, Conclusion

Observation: Normalized artifact records describe device and file activity.

Inference: Temporal relationships are consistent with the generated correlation events.

Conclusion: The combination of device, filesystem, user-session, and application artifacts strongly supports the simulated data-transfer hypothesis. No single artifact independently esta
blishes the complete transfer sequence.

Experimental Score

Score: **13**. This is an experimental project model and is NOT an established forensic standar
d.

Limitations

- Simulated laboratory environment

- Artifact availability depends on Windows configuration

- Timestamp sources have different semantics and reliability

- USB presence alone does not establish transfer

- Correlation does not establish intent

- The experimental score is not a forensic standard

Appendix — Disclaimer

This system is an educational digital-forensics research platform operating on a controlled lab oratory dataset. Findings represent an analytical interpretation of the supplied artifacts and should not be treated as a substitute for a validated forensic methodology in a real investigat ion.